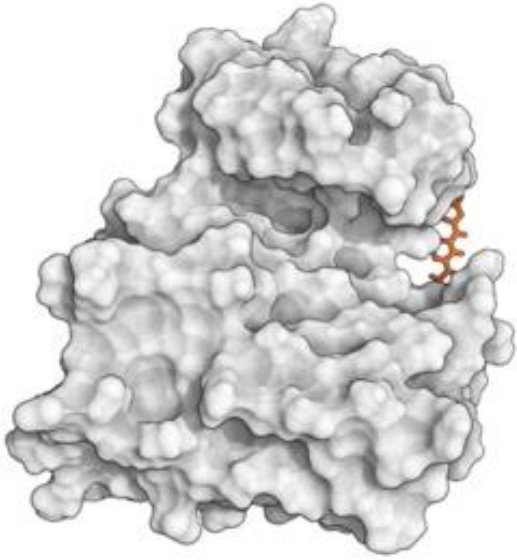


Simulation of Biomolecules



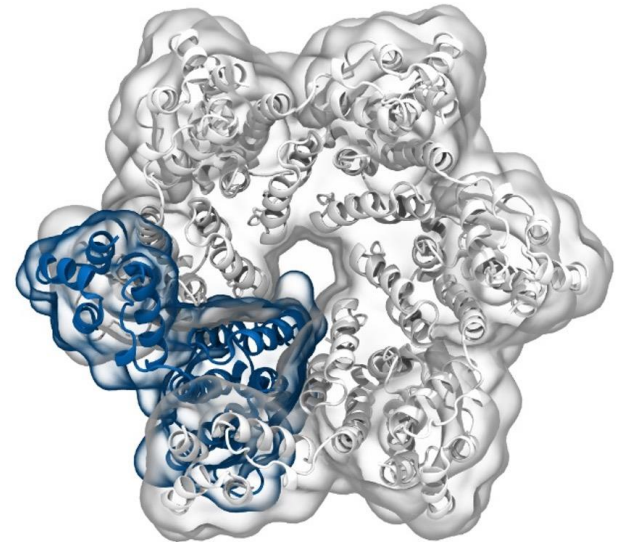
Dr Matteo Degiacomi &
Dr Antonia Mey

University of Edinburgh

matteo.degiacomini@ed.ac.uk

antoniamey@ed.ac.uk

Setting up a protein simulation



Dr Daniel Cole

Newcastle University

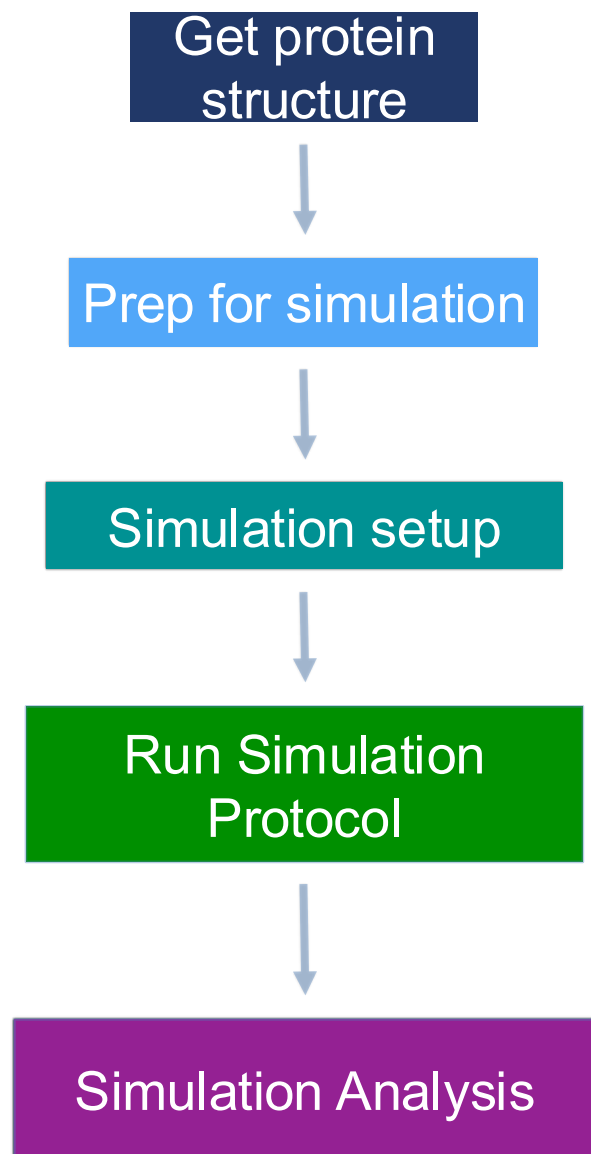
daniel.cole@ncl.ac.uk

Dr Finlay Clark

Newcastle University

finlay.clark@newcastle.ac.uk

A typical workflow for molecular dynamics



Getting your protein structure

AlphaFold Protein Structure Database

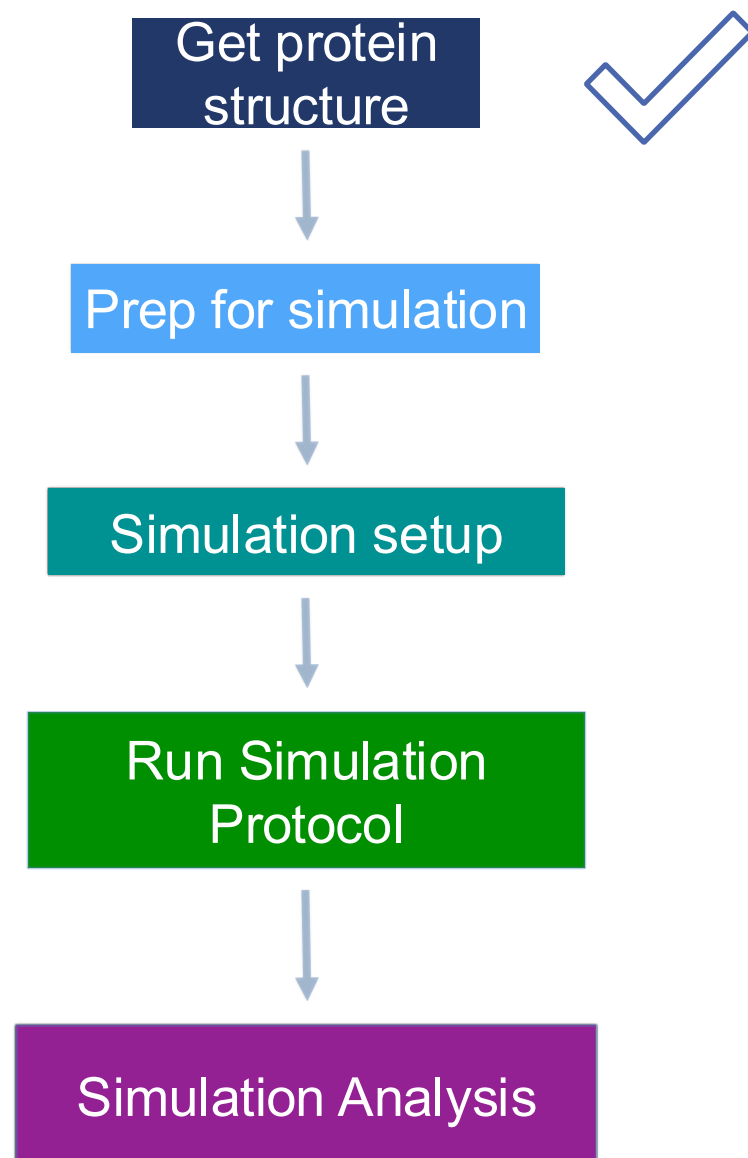


Getting ligands/co-factors

ZINC20



A typical workflow for molecular dynamics



Prepping your protein and/or ligand(s)

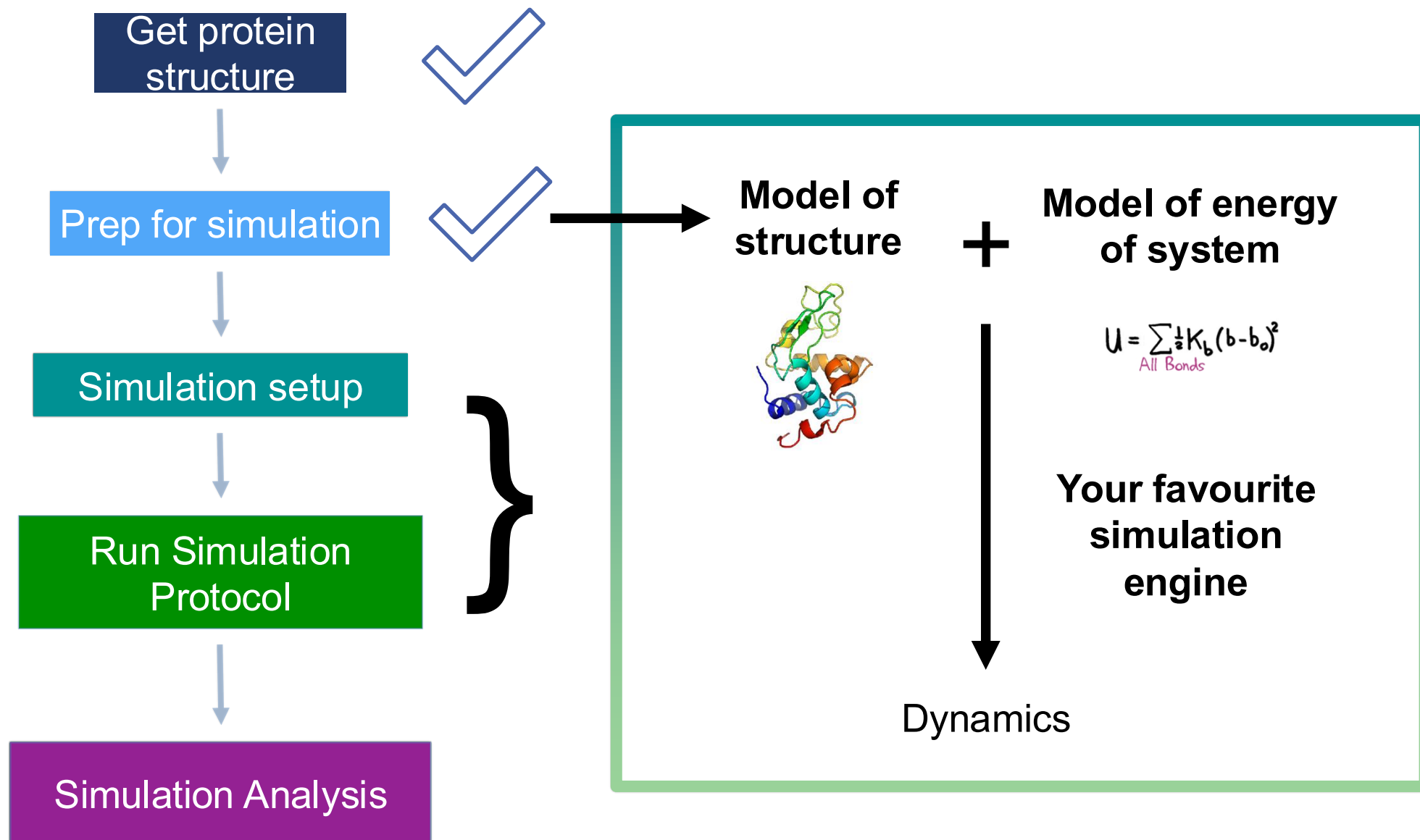
SCHRÖDINGER
Maestro



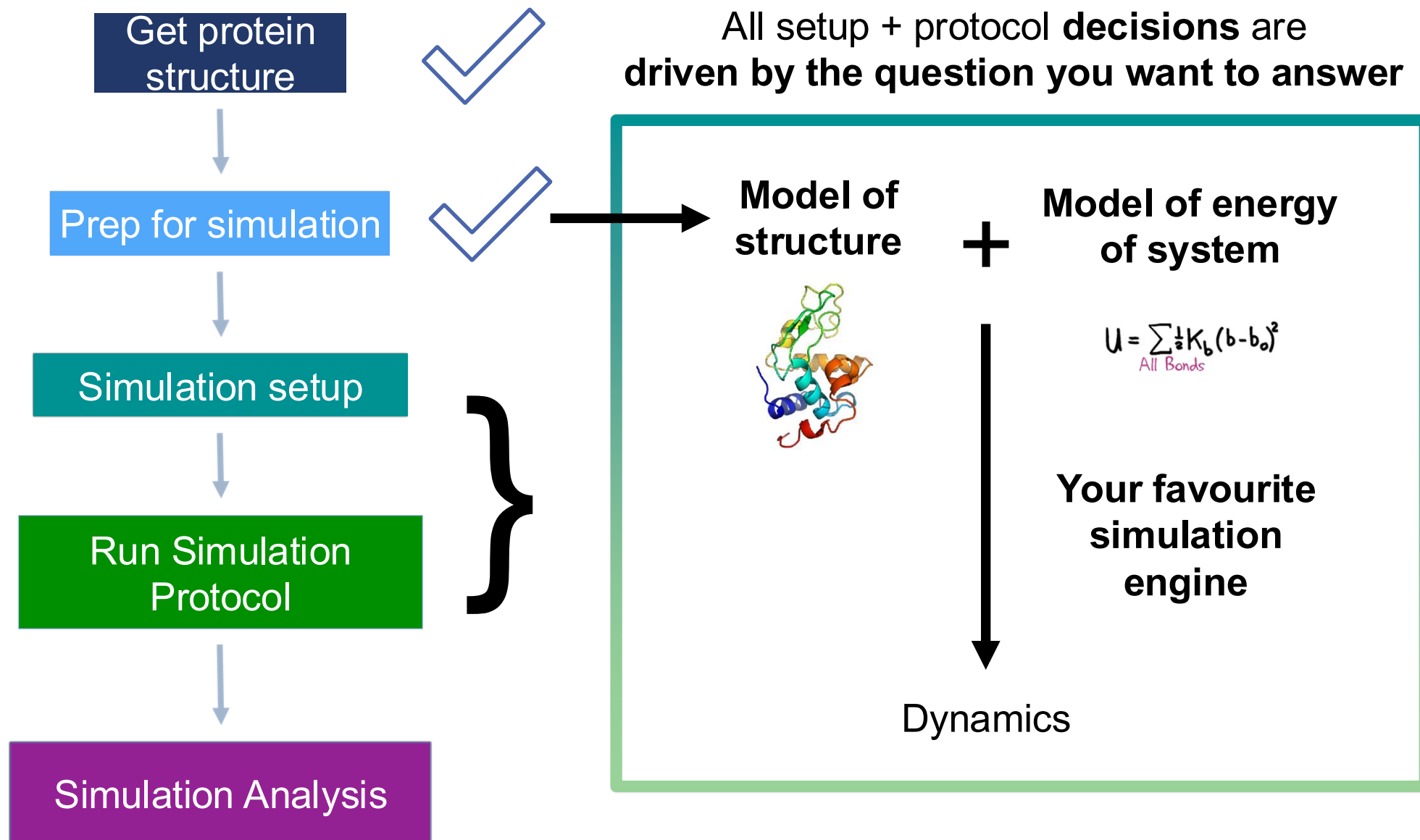
SeeSAR
The drug design dashboard

[...]

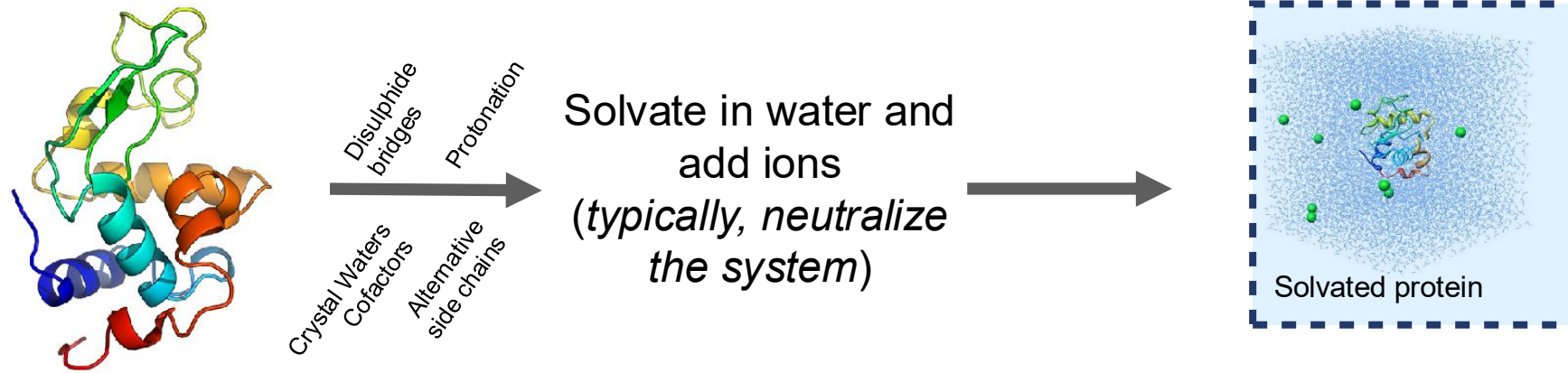
A typical workflow for molecular dynamics



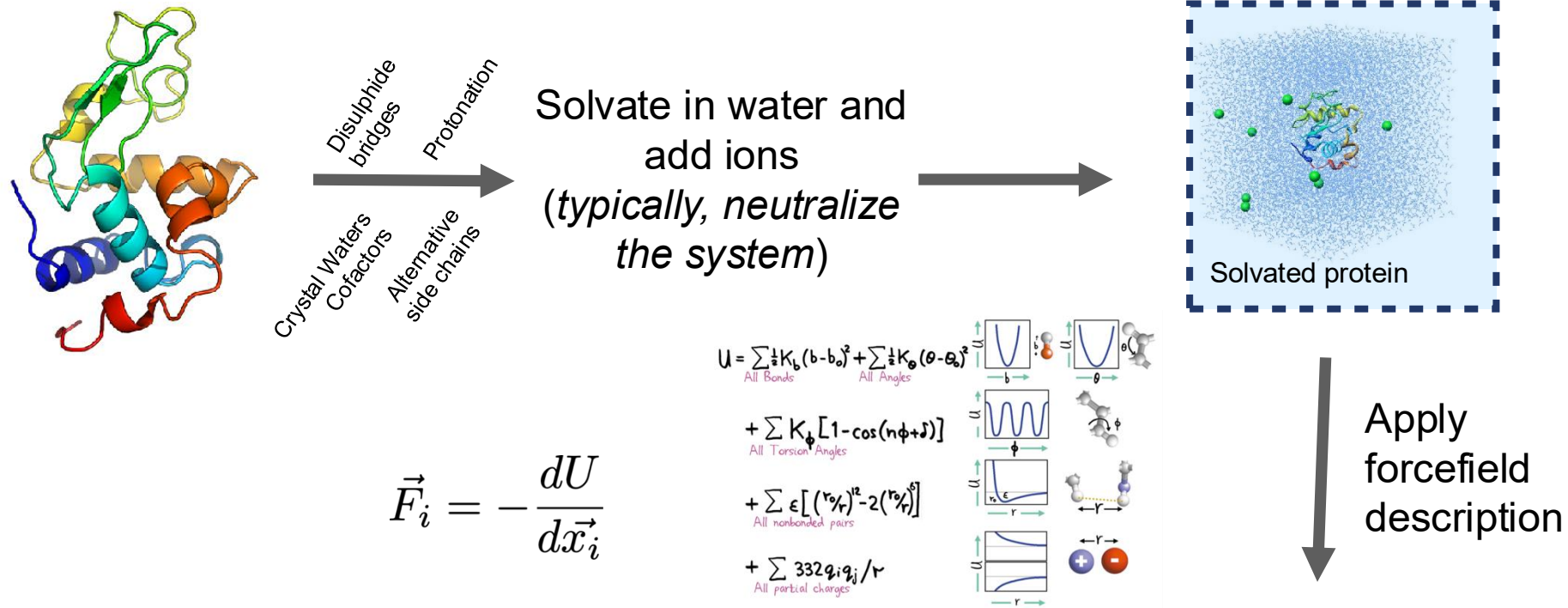
A typical workflow for molecular dynamics



Molecular dynamics require multiple steps for the setup of simulations



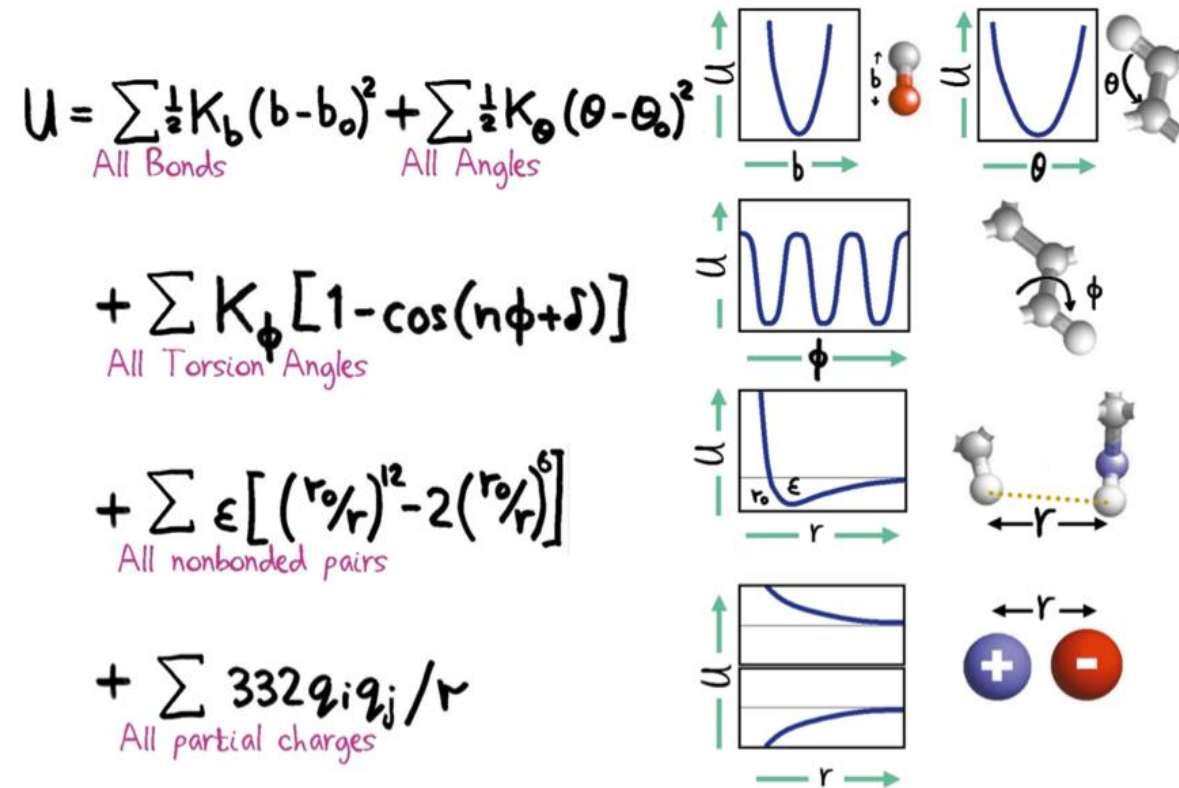
Molecular dynamics require multiple steps for the setup of simulations



Molecular mechanics force fields balance accuracy and speed

- Often interested in relatively long-timescale motions
- Requires fast energy function

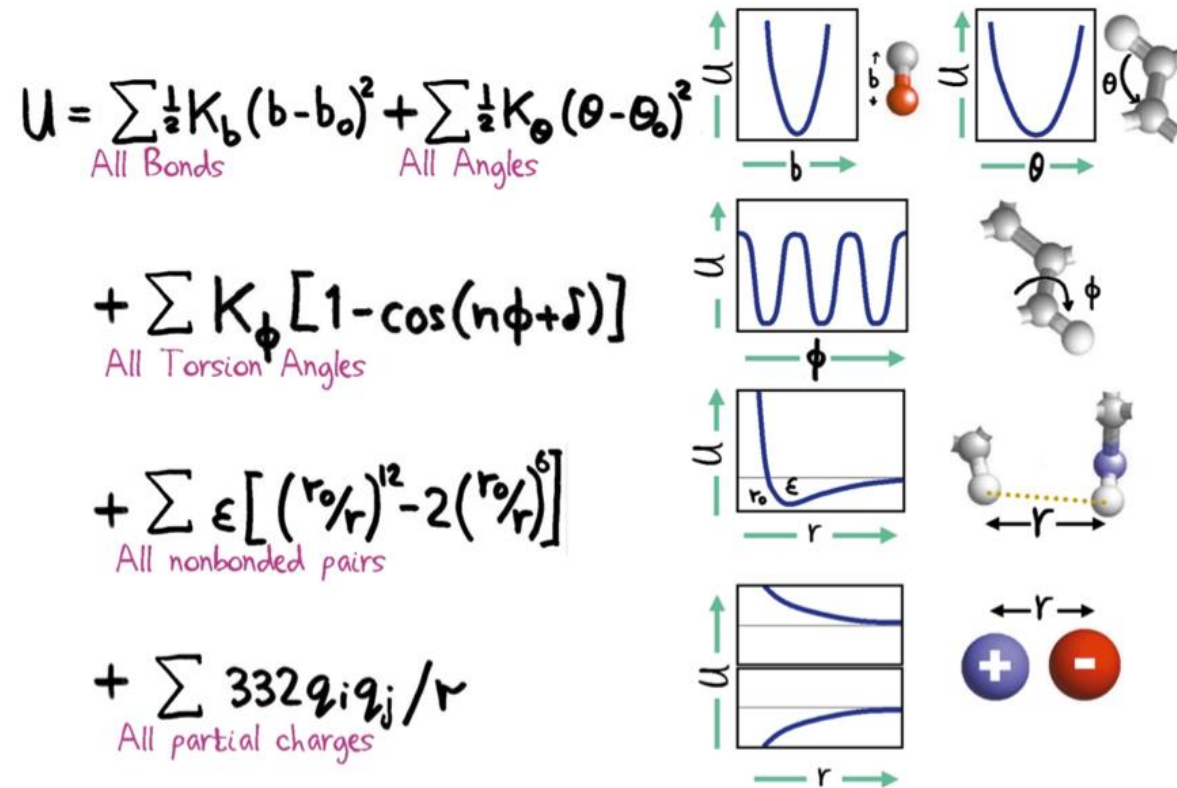
$$\vec{F}_i = - \frac{dU}{d\vec{x}_i}$$



Molecular mechanics force fields balance accuracy and speed

- The functional form + **parameters** decide accuracy

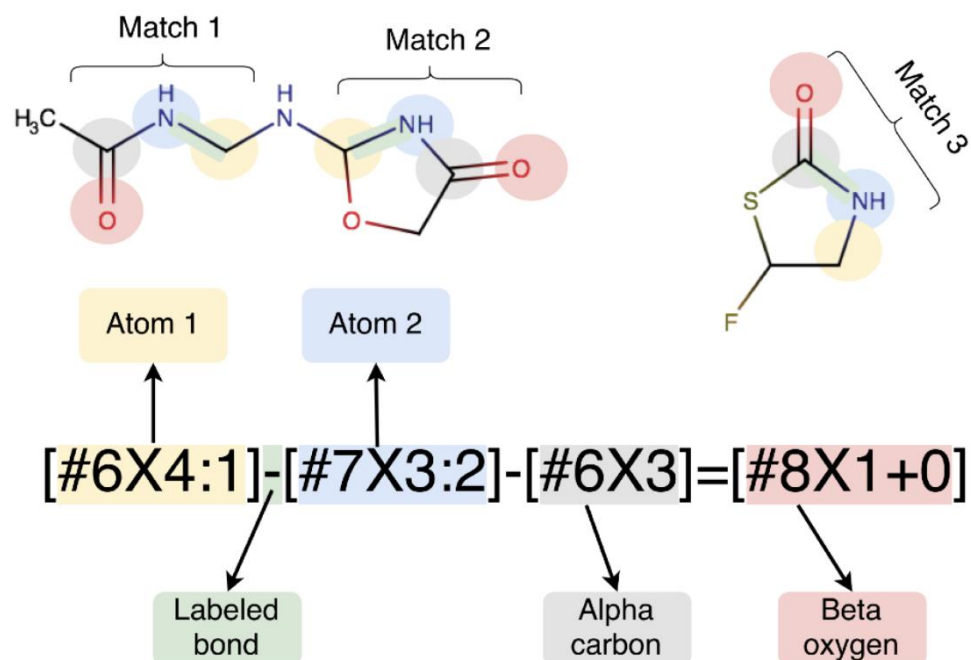
$$\vec{F}_i = - \frac{dU}{d\vec{x}_i}$$



Molecular mechanics force fields balance accuracy and speed

Parameters are assigned using discrete types

**SMIRKS Native
Open Force Field
Specification**



Mobley et al., JCTC, 2018, **14**, 6076

Molecular mechanics force fields balance accuracy and speed

Parameters are assigned using discrete types

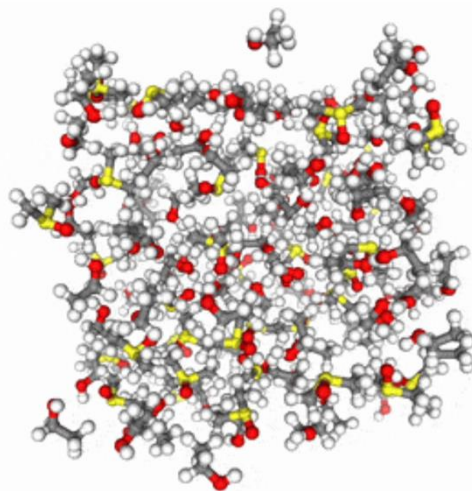
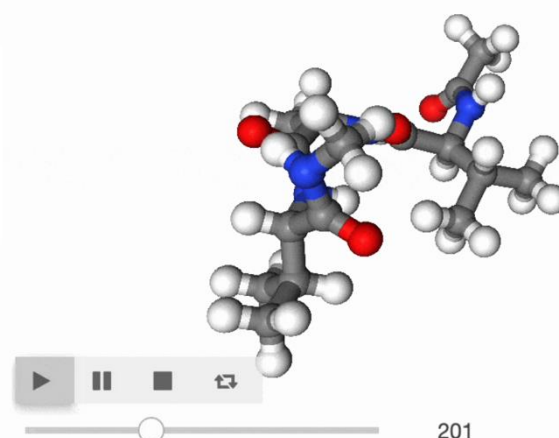
```
<Bonds version="0.4" potential="harmonic" fractional_bondorder_method="AM1-Wiberg" fractional_bondorder_interpolation="linear">  
  <Bond smirks="[#6X4:1]-[#6X4:2]" id="b1" length="1.525970013793 * angstrom ** 1" k="457.9258198725 * kilocalorie_per_mole ** 1
```

$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_0)^2$$

Molecular mechanics force fields balance accuracy and speed

Parameters are fit to QM + experimental data

Bonded parameters trained on curated QM energies and forces of 1000's of drug-like molecules and fragments

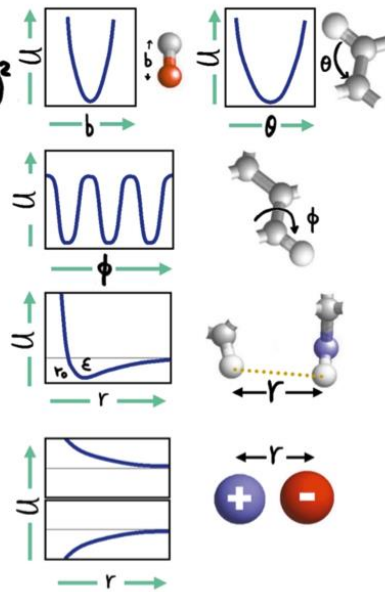


Non-bonded training set includes 1000's of mixture enthalpy and density measurements from NIST ThermoML

Many force fields are available

Biomolecular force fields

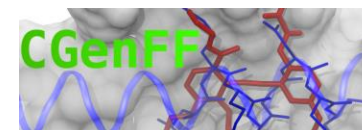
$$\vec{F}_i = - \frac{dU}{d\vec{x}_i}$$

$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_0)^2 + \sum_{\text{All Angles}} \frac{1}{2} K_\theta (\theta - \theta_0)^2 + \sum_{\text{All Torsion Angles}} K_\phi [1 - \cos(n\phi + \delta)] + \sum_{\text{All nonbonded pairs}} \epsilon \left[\left(\frac{r_0}{r} \right)^{12} - 2 \left(\frac{r_0}{r} \right)^6 \right] + \sum_{\text{All partial charges}} \frac{332 q_i q_j}{r}$$


The diagram illustrates the components of a biomolecular force field. It shows four potential energy curves (U vs. coordinate) corresponding to different types of interactions: bonds (U vs. b), angles (U vs. θ), torsions (U vs. φ), and nonbonded pairs (U vs. r). Each curve is accompanied by a small molecular model showing the relevant atoms and their coordinates. The nonbonded interaction curve shows a strong repulsive wall at short distances and a shallow attractive well at longer distances, with a dashed line indicating the equilibrium distance r₀. The partial charge interaction is represented by a simple Coulombic potential between two charges, qᵢ and qⱼ, separated by a distance r.

- **Amber** (incl. Glycam params for glycans)
- **CHARMM** (incl. POPC, POPE, DPPC lipids)
- **OPLS**
- **GROMOS**
- ...

Small molecule force fields



$$E_{\text{pair}} = \sum_{\text{bonds}} K_r (r - r_{eq})^2 + \sum_{\text{angles}} K_\theta (\theta - \theta_{eq})^2 + \sum_{\text{dihedrals}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^6} + \frac{q_i q_j}{\epsilon R_{ij}} \right]$$

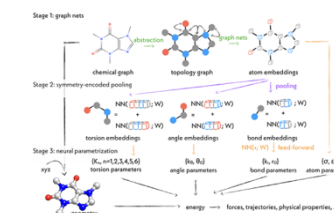
GAFF

Machine learned force fields



Espaloma

Ani-2x



SchNet

PhysNet

BandNN

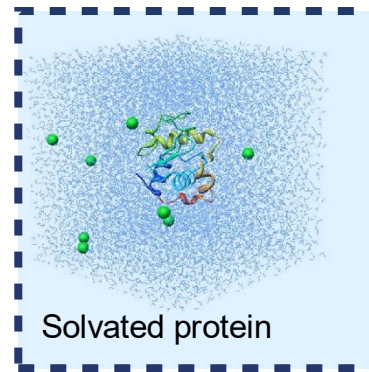
MACE-OFF

There is no “best force field”!

Many force fields are available ... but not all are compatible

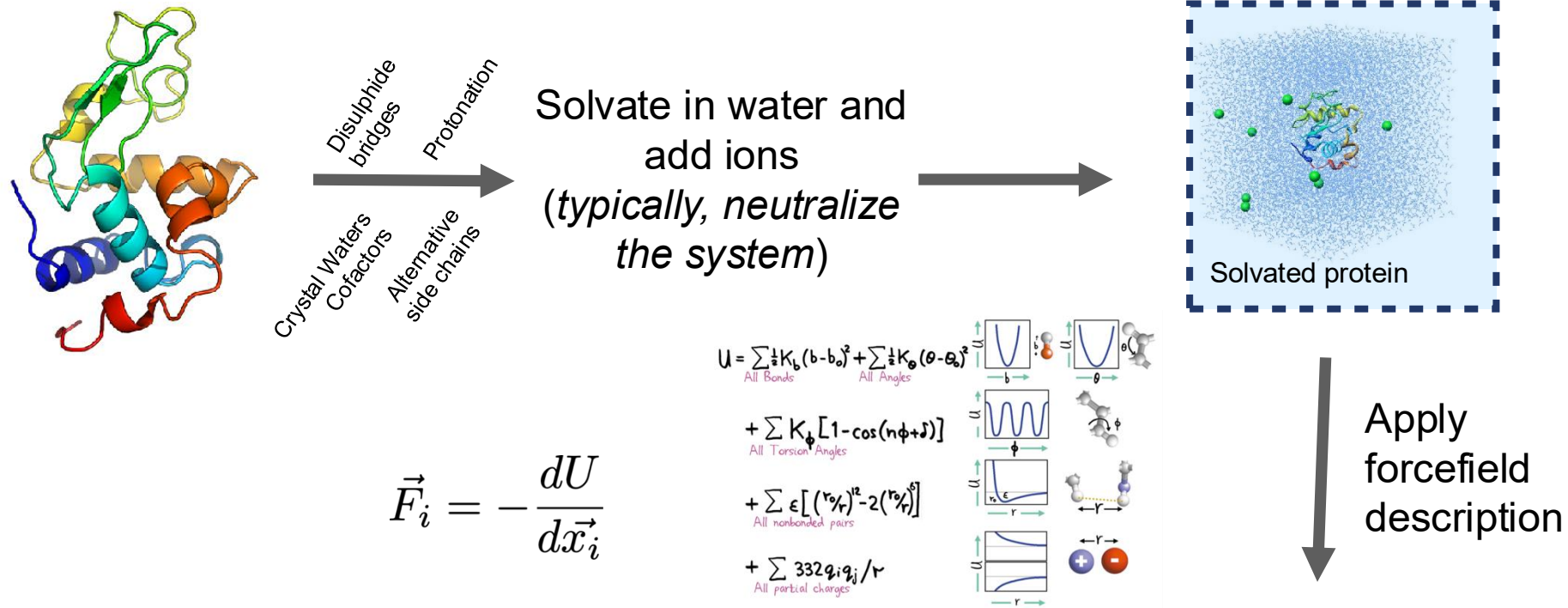
Be careful to use compatible water models and charge models etc...

AMBER ff14SB <=> TIP3P water
AMBER ff19SB <=> OPC water

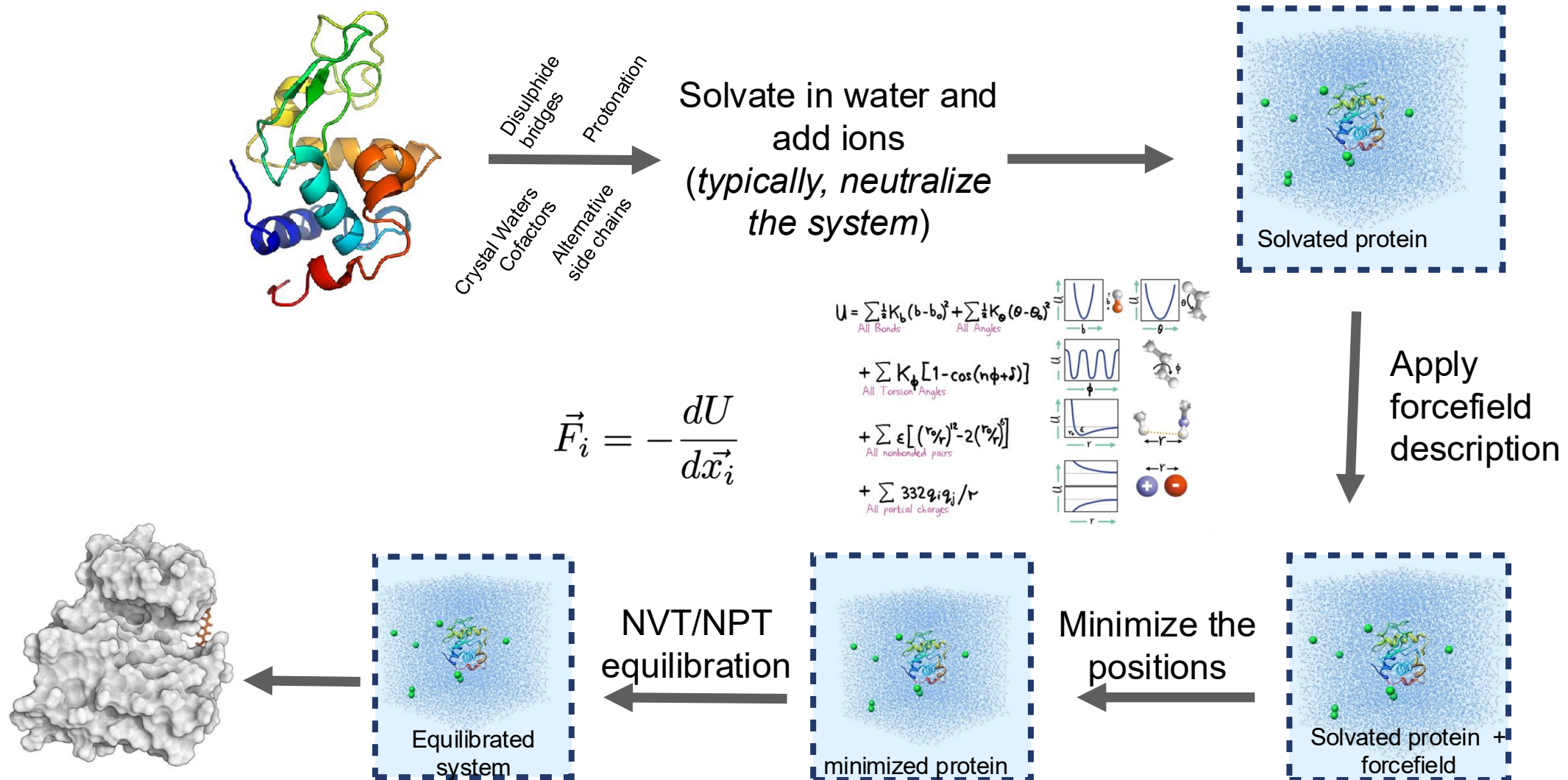


There is no “best force field”!

Molecular dynamics require multiple steps for the setup of simulations



Molecular dynamics require multiple steps for the setup of simulations



Production simulation

Choosing your thermodynamic ensemble

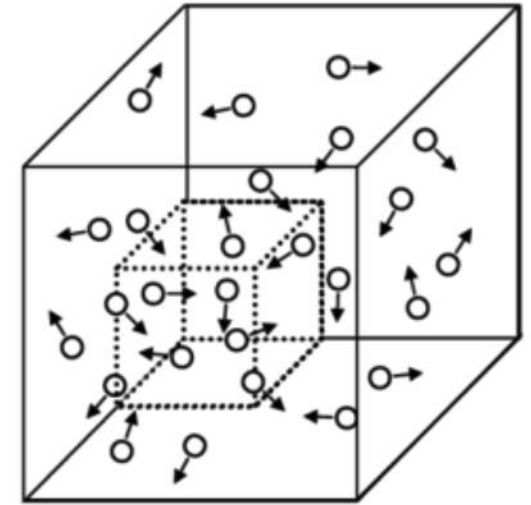
Simulations replicate a specific *thermodynamic ensemble* (typically NVT or NPT), or even grand canonical (μ VT)

You will have different options to include *thermostats* (scaling atom velocities) and *barostats* (scaling positions) in your calculations:

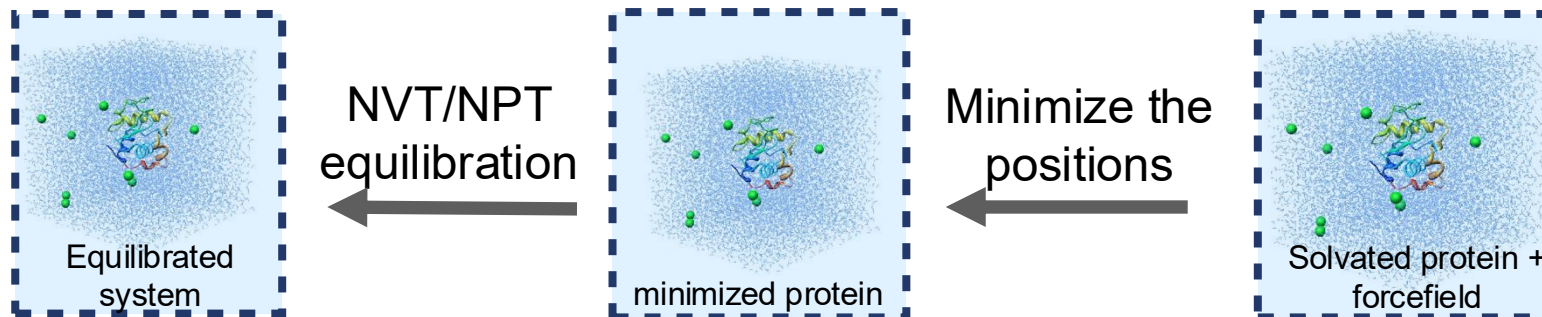
- Nose-Hoover
- Berendsen
- Parrinello-Rahman
- Langevin piston
- ...

Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun¹, Justin Gilmer², Heather B. Mayes³, David L. Mobley⁴, Jacob I. Monroe⁵, Samarjeet Prasad⁶, Daniel M. Zuckerman⁷

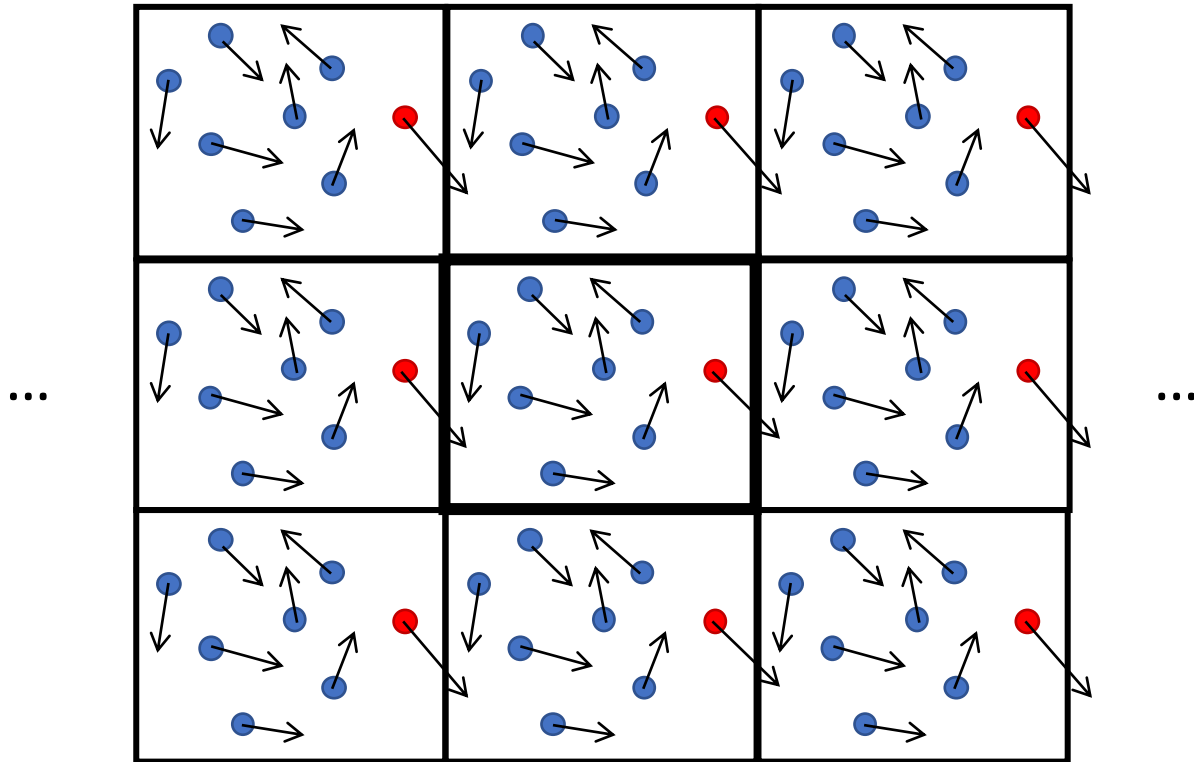


μ VT

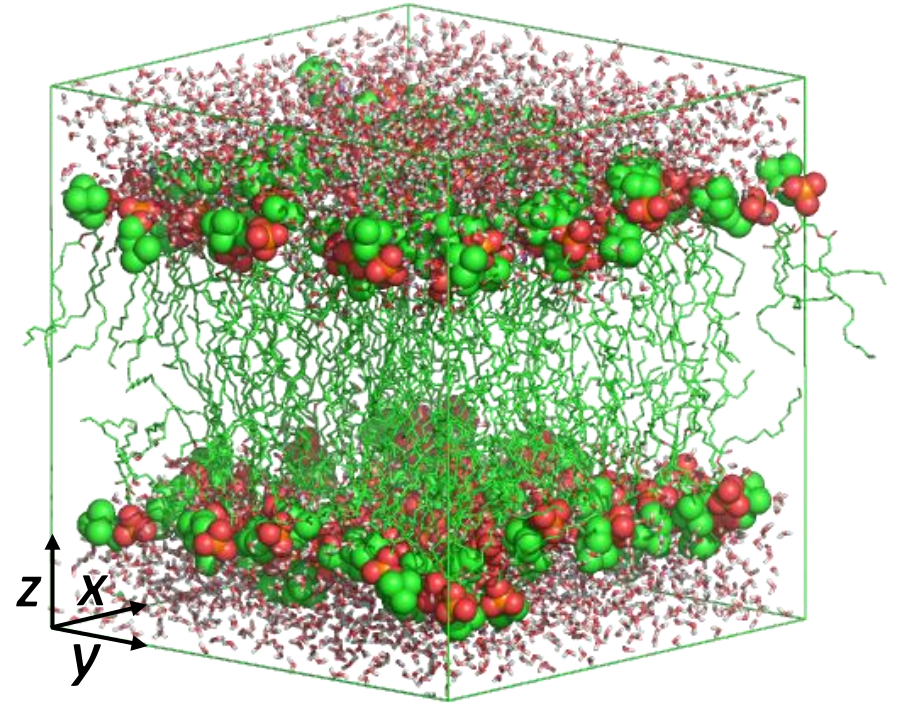


Periodic boundary conditions (PBC) and pressure coupling

Useful to reduce finite-size effect and simulate bulk



Typically, PBC applied in x, y and z direction



For membrane systems, use semi-isotropic pressure coupling ($x, y \neq z$, lipids compressibility is direction-dependent)

Sampling timescales for protein systems

The fastest vibration determines the smallest timestep.

Timestep size is imposed by the fastest phenomenon we want to observe.

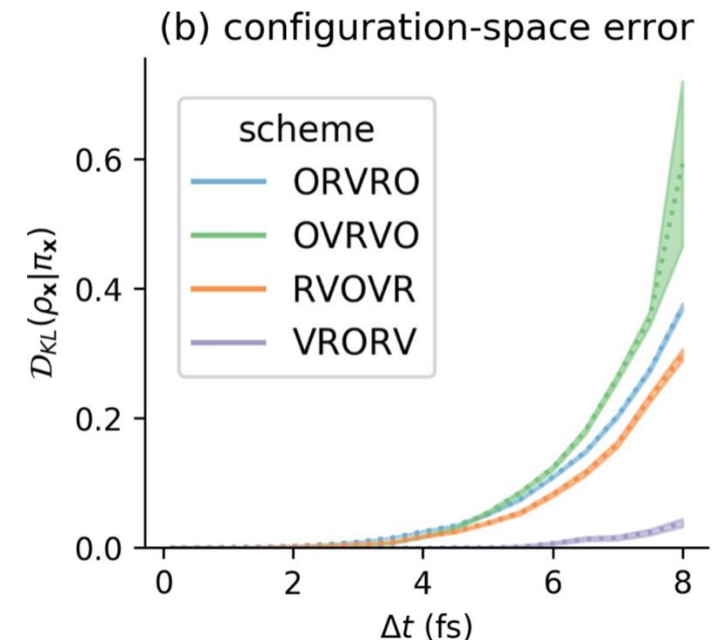
In atomistic simulations:

- Covalent bond hydrogen-heavy atom (10^{14} Hz): 0.5 fs
- Covalent bond heavy atom-heavy atom: 1 fs
- Angles fluctuations: 2 fs

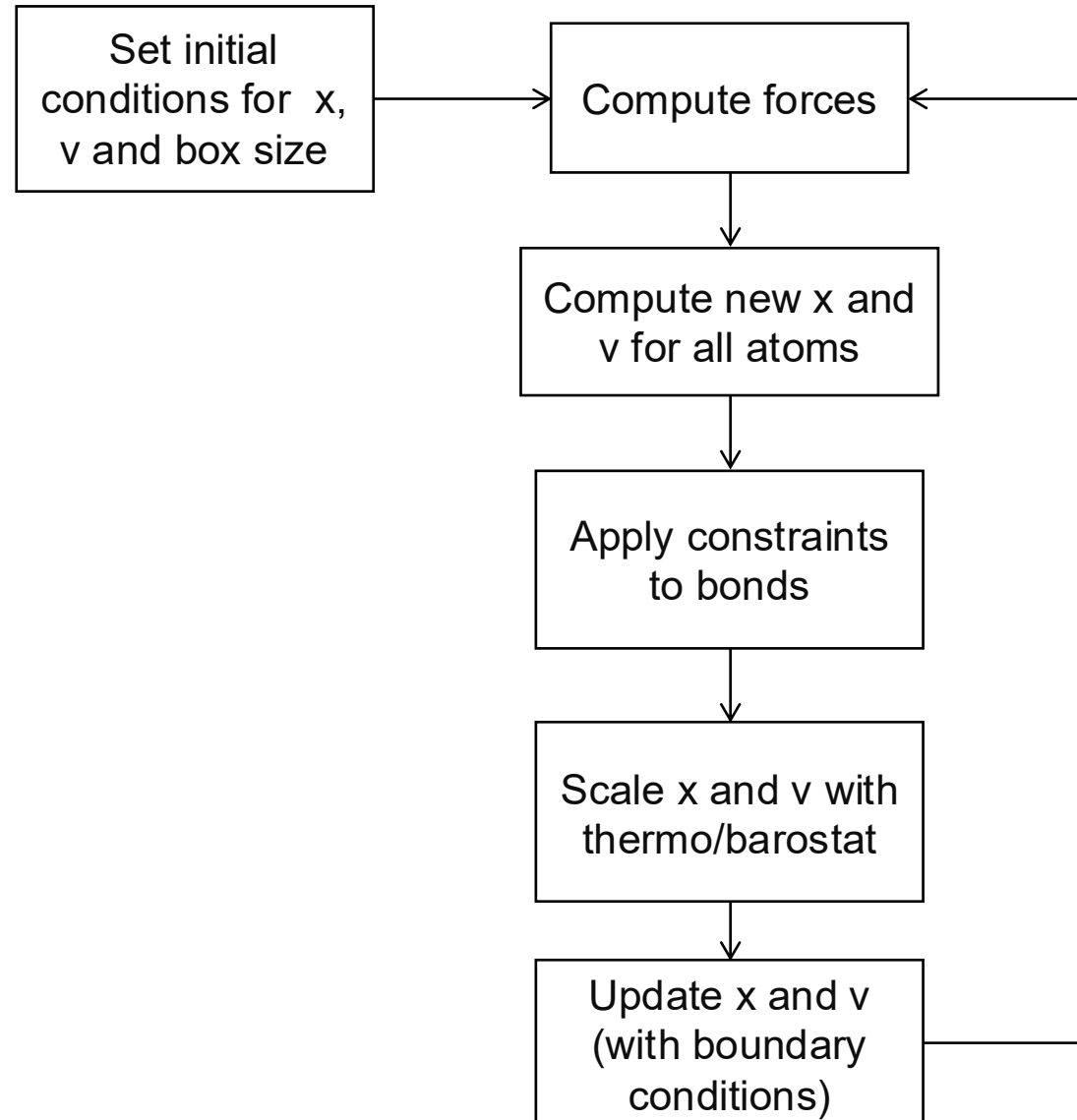
Restraining covalent bond distances allows to use 1-2 fs timesteps (restraining methods: SHAKE, RATTLE, LINCS,...)

Hydrogen Mass repartitioning: 4 fs

Other integrators (e.g., Langevin): 4 fs - 6 fs.



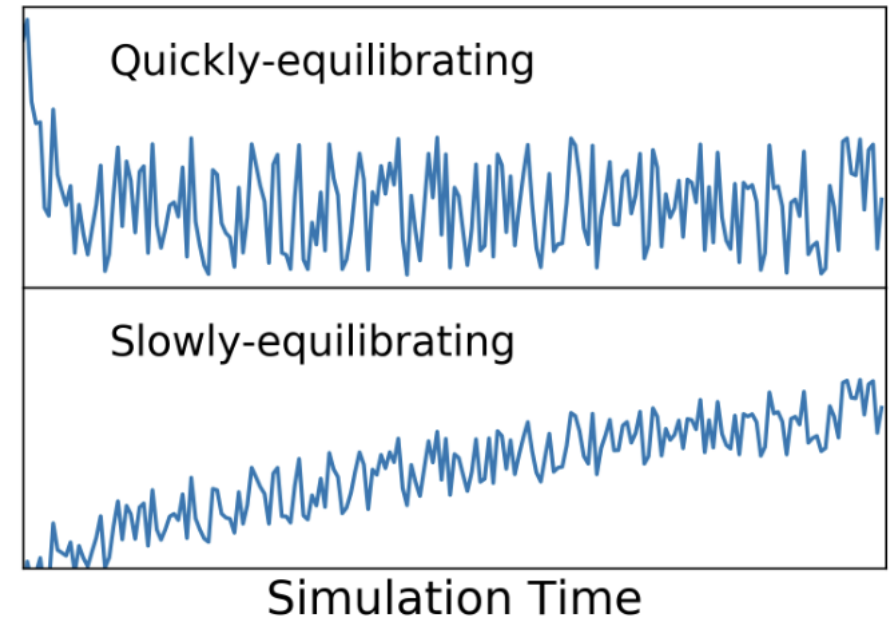
A Molecular Dynamics timestep



“equilibration” and “convergence”: what do they mean?

“Equilibration”:

- Relaxation from starting configuration which has very low equilibrium probability in your ensemble
- “Unequilibrated” data usually discarded
- Related to when you start collecting data

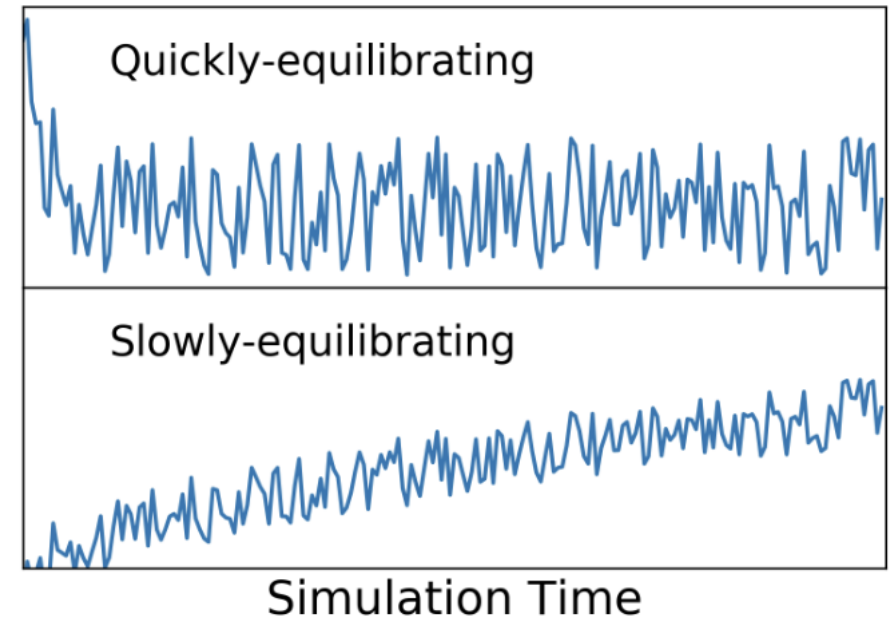


<https://doi.org/10.33011/livecoms.1.1.5957>

“equilibration” and “convergence”: what do they mean?

“Convergence”:

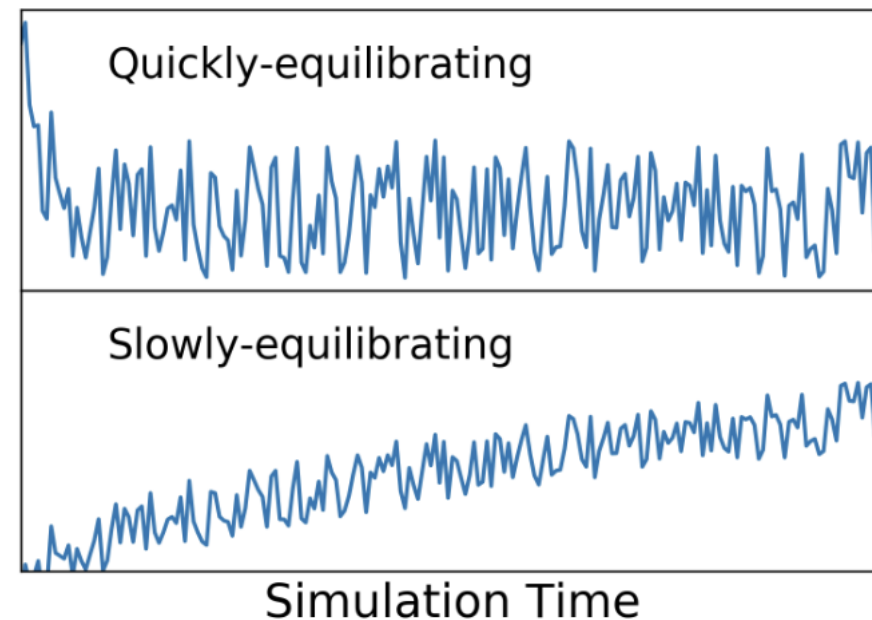
- Extent to which an estimator approaches some “true” value (expectation of observable) with increasing amounts of data
- Often approximated by agreement of replicates
- Related to when you stop collecting your data



<https://doi.org/10.33011/livecoms.1.1.5957>

“equilibration” and “convergence”: what do they mean?

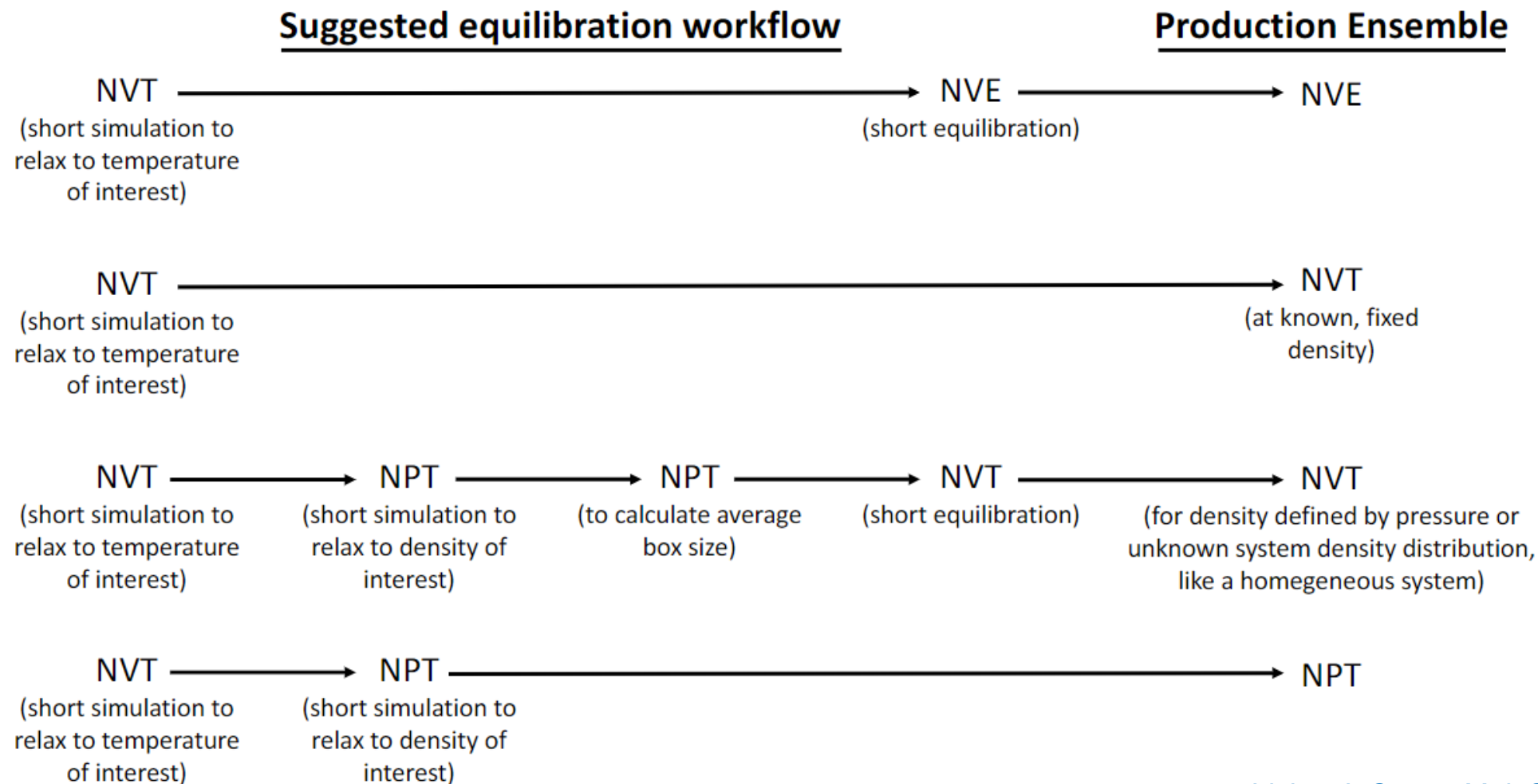
- Different quantities will “equilibrate” and “converge” at different rates
- Sometimes obvious we have not “converged/ equilibrated”, but almost never that we have
- Running independent replicates is best practice for uncertainty quantification



<https://doi.org/10.33011/livecoms.1.1.5957>

Example equilibration protocols

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)



An *example* simulation protocol

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)

Equilibration:

1. Minimize energy, 1000 steepest descent
2. Heat system from 0 to 300 K in 500 ps, NPT, Berendsen barostat 1 atm. α -carbon restrained with 10 kcal/mol harmonic potential. 2 fs timestep, LINCS all bonds
3. 1 ns NVT equilibration with Langevin dynamics, no atom constrained.

Production:

4. 1 μ s NPT, Nose-Hoover barostat, PME for electrostatics

DETERMINE HANDLING OF CUTOFFS

- ☐ As a general rule, electrostatics are long-range enough that either the cutoff needs to be larger than the system size (for finite systems) or periodicity is needed along with full treatment of long-range electrostatics (Section 3.4)
- ☐ Nonpolar interactions can often be safely treated with cutoffs of 1-1.5 nm as long as the system size is at least twice that, but long-range dispersion corrections may be needed (Section 4.1)

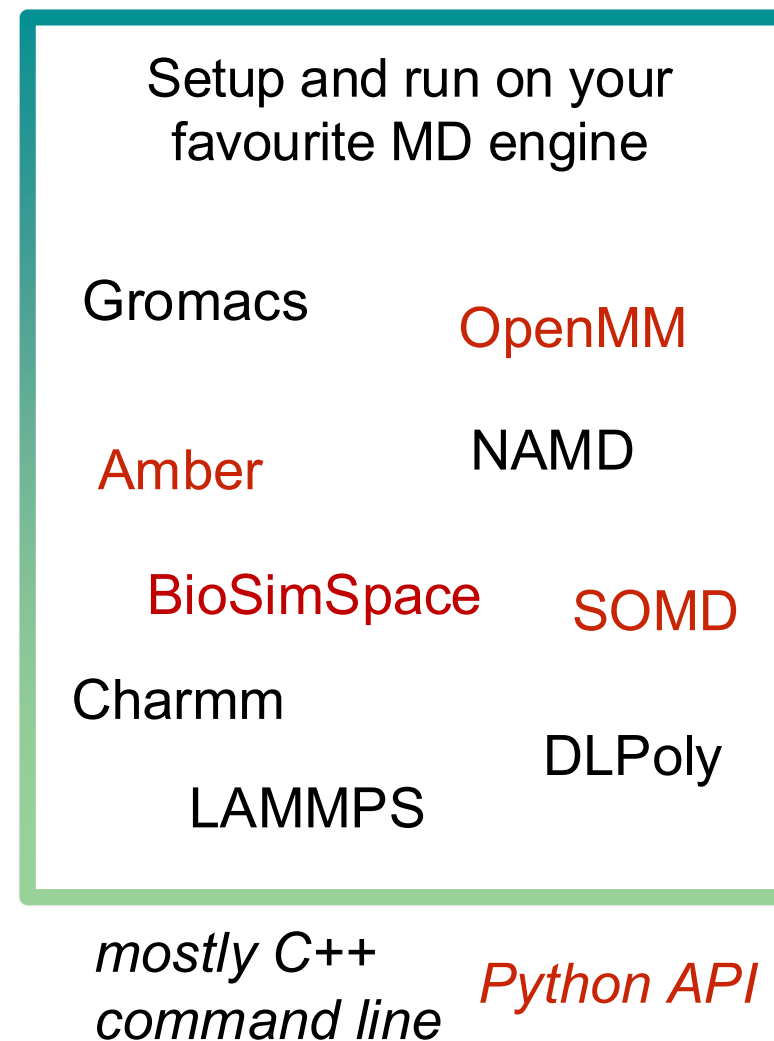
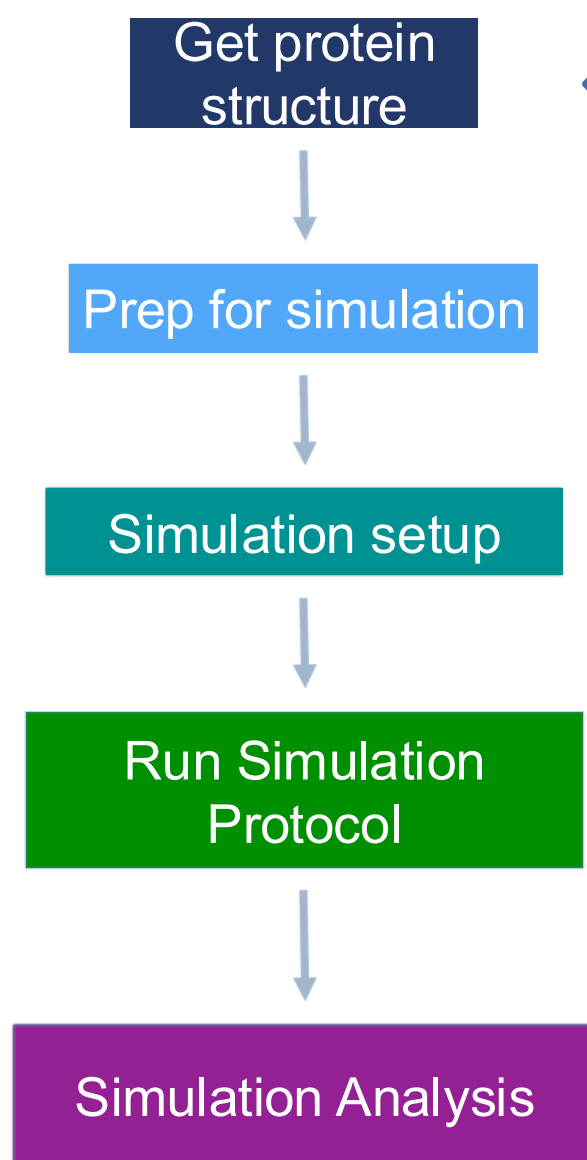
CHOOSE APPROPRIATE SETTINGS FOR THE DESIRED ENSEMBLE

- ☐ Pick a thermostat that gives the correct distribution of temperatures, not just the correct average temperature; if you have a small system or a system with weakly interacting component choose one which works well even in the small-system limit.
- ☐ Pick a barostat that gives the correct distribution of pressures
- ☐ Consider the known shortcomings and limitations of certain integrators and thermostats/barostats and whether your choices will impact the properties you are calculating

CHOOSE AN APPROPRIATE TIMESTEP FOR STABILITY AND AVOIDING ENERGY DRIFT

- ☐ Determine the highest-frequency motion in the system (typically bond vibrations unless bond lengths are constrained)
- ☐ As a first guess, set the timestep to approximately one tenth of the highest-frequency motion's characteristic period
- ☐ Test this choice by running a simulation in the microcanonical ensemble, and ensure that energy is conserved

A typical workflow for molecular dynamics



Which MD engine should I use?

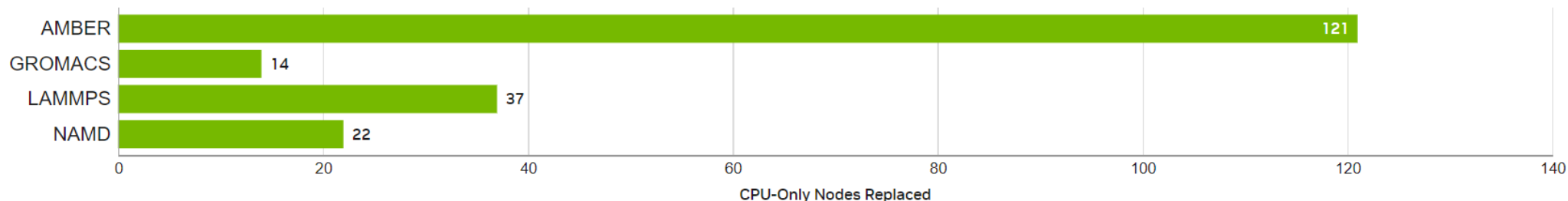
Consider:

- Support for force field of choice
 - Enables running desired simulation protocol
 - Performance for available hardware
 - Ease of use
- depends on number of atoms, hardware, simulation protocol, MD engine*

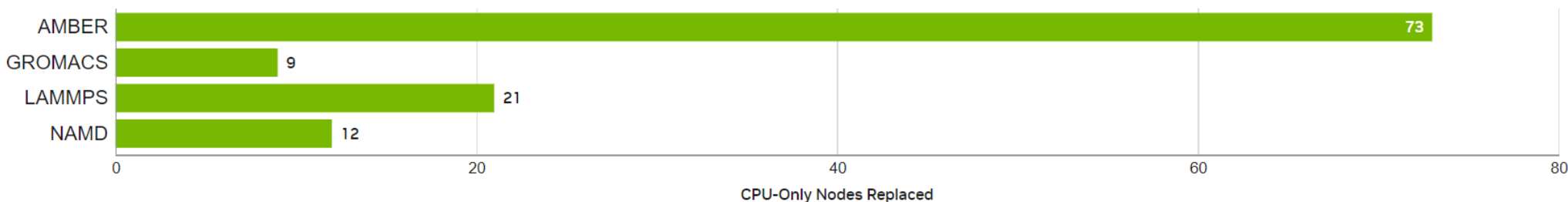
Graphical Processing Units (GPUs) are especially effective for MD

From: developer.nvidia.com/hpc-application-performance

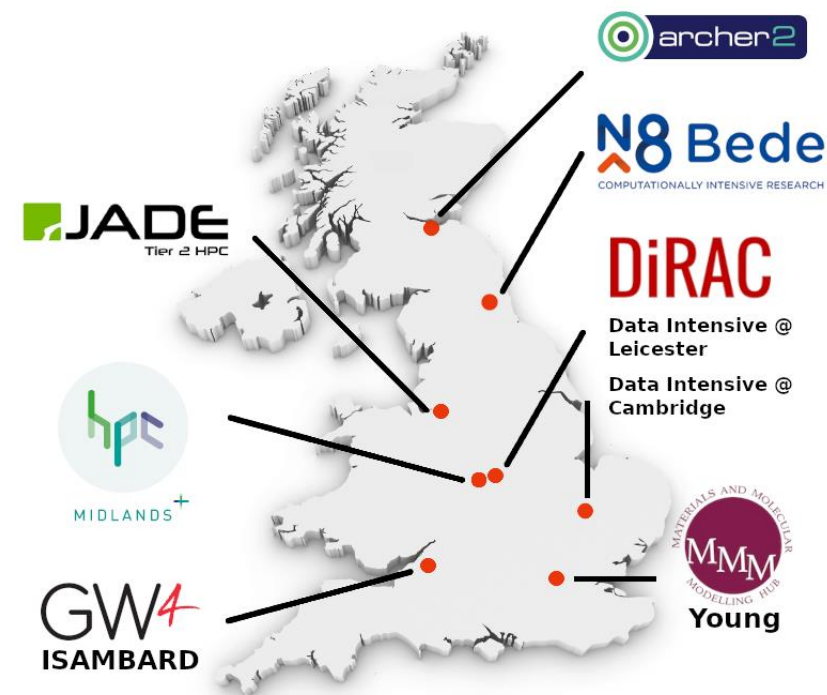
NVIDIA H200



NVIDIA A100



Calculating runtimes: example on UK Tier 2 systems



[as of 2024]

HECBioSim

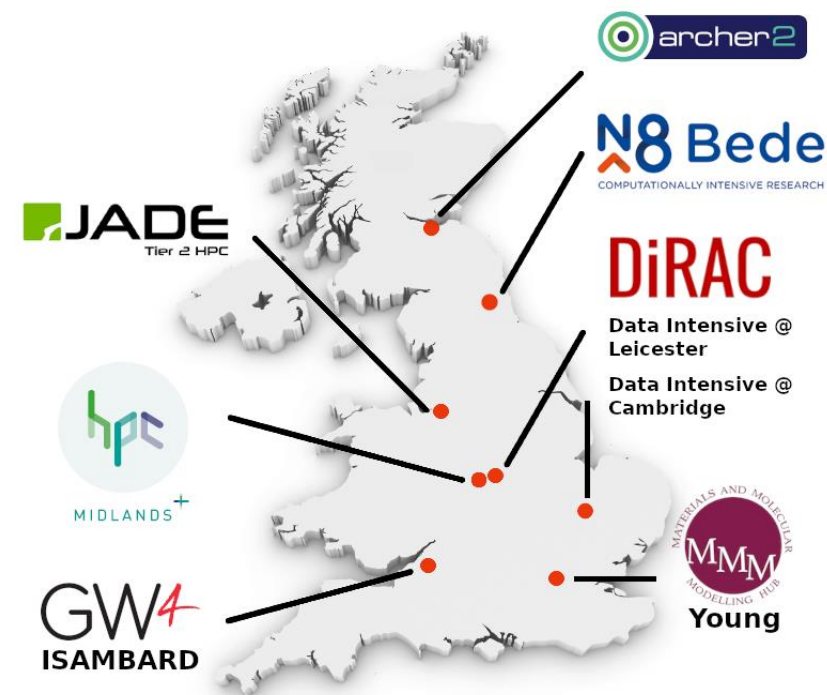
Search ...

HPC Calculator

HPC System:	Software:	No. of atoms to simulate:	Simulation time (ns):
JADE2	OpenMM	50000	500

<https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator>

Calculating runtimes: example on UK Tier 2 systems



[as of 2024]

 **HECBioSim**

Search ...



HPC Calculator

HPC System:
JADE2

Software:
OpenMM

No. of atoms to simulate:
50000

Simulation time (ns):
500

🕒 179 ns/day

💾 47.5GB of storage

🕒 67 node hours

👤 1 node recommended for maximum efficiency

⚡ 33.2 kWh of power

🏠 ...which could power an average UK home for 4.49 days

<https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator>

Disclaimer!



Running biomolecular MD can take days on specialised hardware.

Today we will *not* run any of them, and instead will focus on fundamental principles using small molecules.

Workshop 4: OpenFF Toolkit + Interchange



OPEN SOFTWARE

Automated infrastructure enables rapid experimentation with minimum human intervention



OPEN DATA

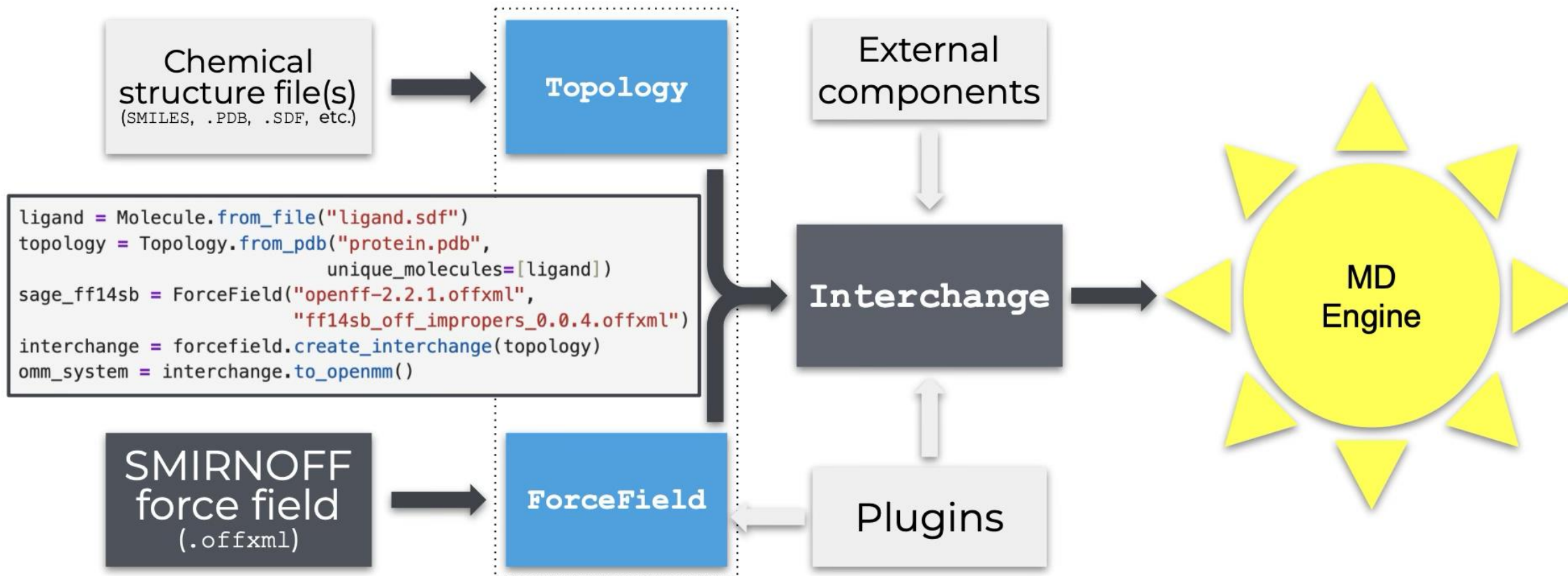
Access to large, high quality experimental and quantum chemical data facilitates easy curation of balanced train / test sets



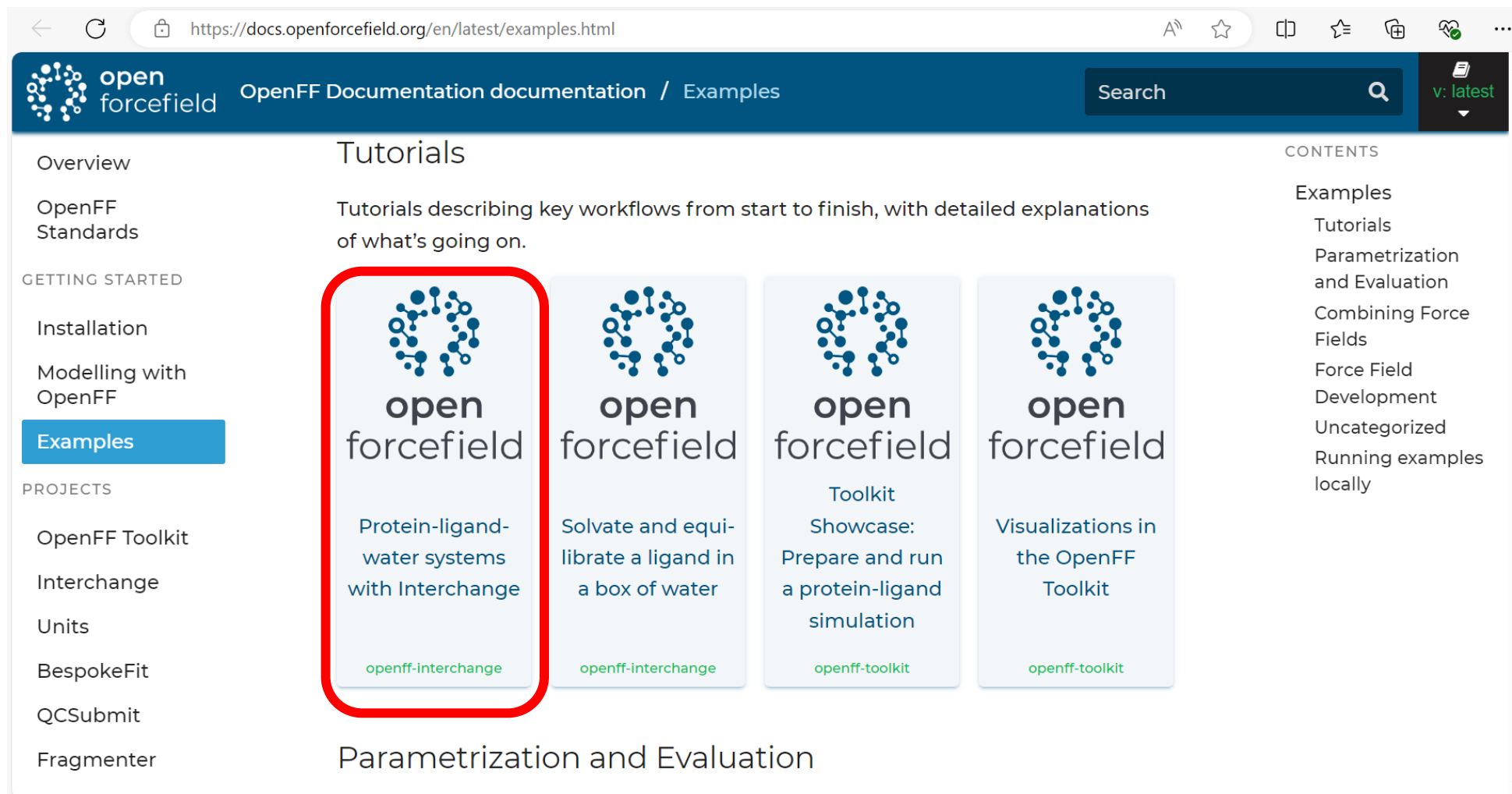
OPEN SCIENCE

Exploring new force field science:
hypothesis - build
software - train - test -
iterate
is now almost routine

Workshop 4: OpenFF Toolkit + Interchange



Workshop 4: OpenFF Toolkit + Interchange

A screenshot of the OpenFF Documentation website. The browser address bar shows "https://docs.openforcefield.org/en/latest/examples.html". The page has a dark blue header with the OpenFF logo, "OpenFF Documentation documentation / Examples", a search bar, and a version selector set to "v: latest". On the left, a sidebar lists navigation options: Overview, OpenFF Standards, GETTING STARTED (Installation, Modelling with OpenFF, Examples), PROJECTS (OpenFF Toolkit, Interchange, Units, BespokeFit, QCSubmit, Fragmenter). The main content area is titled "Tutorials" and describes "Tutorials describing key workflows from start to finish, with detailed explanations of what's going on." Below this are four tutorial cards, each with the OpenFF logo and a title. The first card, "Protein-ligand-water systems with Interchange", is highlighted with a red rounded rectangle and has the tag "openff-interchange" at the bottom. The second card is "Solvate and equilibrate a ligand in a box of water" with tag "openff-interchange". The third card is "Toolkit Showcase: Prepare and run a protein-ligand simulation" with tag "openff-toolkit". The fourth card is "Visualizations in the OpenFF Toolkit" with tag "openff-toolkit". On the right, a "CONTENTS" sidebar lists "Examples" (Tutorials, Parametrization and Evaluation, Combining Force Fields, Force Field Development, Uncategorized, Running examples locally).

<https://docs.openforcefield.org/en/latest/examples.html>

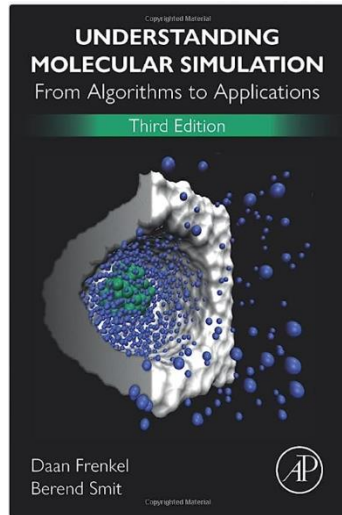
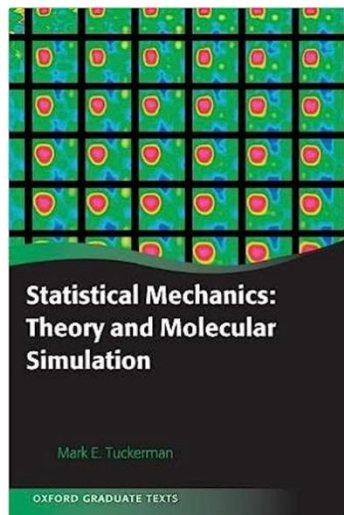
Useful resources to learn running simulations

Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun¹, Justin Gilmer², Heather B. Mayes³, David L. Mobley⁴, Jacob I. Monroe⁵, Samarjeet Prasad⁶, Daniel M. Zuckerman⁷

A suite of tutorials for the BioSimSpace framework for interoperable biomolecular simulation [Article v1.0]

Lester O. Hedges^{1,2*}, Sofia Bariami^{3†}, Matthew Burman², Finlay Clark³, Benjamin P. Cossins⁴, Adele Hardie³, Anna M. Herz³, Dominykas Lukauskis⁵, Antonia S.J.S. Mey³, Julien Michel^{2,3*}, Jenke Scheen^{3‡}, Miroslav Suruzhon⁴, Christopher J. Woods¹, Zhiyi Wu⁴



From Proteins to Perturbed Hamiltonians: A Suite of Tutorials for the GROMACS-2018 Molecular Simulation Package [Article v1.0]

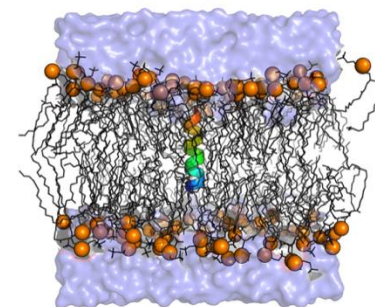
Justin A. Lemkul

Department of Biochemistry, Virginia Polytechnic Institute and State University

<https://orcid.org/0000-0001-6661-8653>

DOI: <https://doi.org/10.33011/livecoms.1.1.5068>

Keywords: tutorials, gromacs, molecular dynamics simulation, computational chemistry



PDF

ARTICLE CODE REPOSITORY

GROMACS: tutorials.gromacs.org

Amber: ambermd.org/tutorials

OpenMM: <https://openmm.github.io/openmm-cookbook/dev/index.html>